

SERAG ELDEIN MAHMOUD ABDULKARIM

Backend Software Engineer | NestJS | Node.js | TypeScript | PostgreSQL | AWS

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

SUMMARY

Backend Software Engineer with hands-on experience building production-ready web applications using NestJS, Node.js, TypeScript, PostgreSQL, MongoDB, Redis, and AWS. Built and deployed multiple client projects as a freelance developer, including payment integrations, authentication systems, AI-powered applications, and scalable REST APIs. Strong foundation in software engineering principles, data structures, algorithms, and system design.

EXPERIENCE

Freelance Software Developer — Self-Employed (Remote)

Jan 2024 – Present

Independently designed, built, and shipped client web applications end to end.

- Built and deployed 5+ full-stack web applications using NestJS, React, PostgreSQL, MongoDB, and AWS.
- Modeled relational and document databases (PostgreSQL, MongoDB) for performance and scalability across client projects.
- Implemented secure payment processing flows using Paymob for client e-commerce and course-platform projects.

AI & ML Intern — IMPACT – Zewail University

Jan 2023 – Sep 2023

- Built an AI-driven document processing system using Python, integrating OpenAI models via REST APIs for automated data extraction.
- Gained hands-on exposure to cloud deployment and database-backed AI pipelines.
- Applied this experience directly to graduation project DocTalker (see Projects).

National Military Service — Egyptian Armed Forces

Jul 2025 – Sep 2026

PROJECTS

Englishhome — Online Courses Platform

<https://englishhome.com>

Tech: NestJS, TypeScript, PostgreSQL, Prisma ORM, AWS S3, AWS SES, Paymob, Passport OAuth

- Built a NestJS + TypeScript backend with PostgreSQL and Prisma ORM using the barrel pattern for maintainability.
- Integrated AWS SES OTP authentication, Google/Facebook OAuth via Passport, and Paymob for payments.
- Used AWS S3 with CDN delivery for course content storage.

DocTalker — AI Document Chat Platform

github.com/itzSerag/DocTalker

Tech: Node.js, Express.js, MongoDB, OpenAI API, AWS S3, Google OAuth

- Built an Express.js API integrating OpenAI, AWS, and MongoDB to power AI-driven document chat.
- Implemented OTP-based and Google OAuth authentication for secure access.
- Engineered voice-to-text input and a document processing pipeline that extracts and summarizes content across formats.

Chatty — Real-Time Chat Application

github.com/itzSerag/realtime_MERN_chatty-backend

Tech: NestJS, MongoDB, Socket.IO, Cloudinary, React, Tailwind CSS

- Implemented the repository pattern in a NestJS backend with MongoDB/Mongoose for clean data-access separation.
- Built real-time messaging using Socket.io and integrated Cloudinary for optimized media storage.
- Built a responsive React + Tailwind/DaisyUI frontend with Zod-validated forms via React Hook Form.

SKILLS

Languages: JavaScript, TypeScript, Python, C++, Java

Backend: Node.js, Express.js, NestJS, Spring Boot, REST API design, Microservices, System design

Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM

Frontend: React.js, Tailwind CSS, MUI

Cloud & DevOps: AWS (S3, EC2, SES), Docker, Git, GitHub Actions

Engineering Practices: Data Structures & Algorithms, 250+ LeetCode Problems Solved, OOP, SOLID Principles, Design Patterns, REST API Design, SDLC, Payment Gateway Integration, Database Design

EDUCATION

B.Sc. in Computer Science — Ain Shams University, Cairo, Egypt

Sep 2020 – Sep 2024

- Coursework: Databases, Algorithms & Data Structures, OOP, Parallel Programming, AI, Java, C++

ECPC — Egyptian Collegiate Programming Contest

Aug 2022

- Competed in the Egyptian Collegiate Programming Contest, applying problem-solving skills in algorithms and data structures under competition conditions.

CERTIFICATIONS

- IBM Web Development Training — [Credly Badge](#)
- IBM Software Developer Development Training — [Credly Badge](#)
- Rest of the Certifications [Certification - Drive](#)

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency)